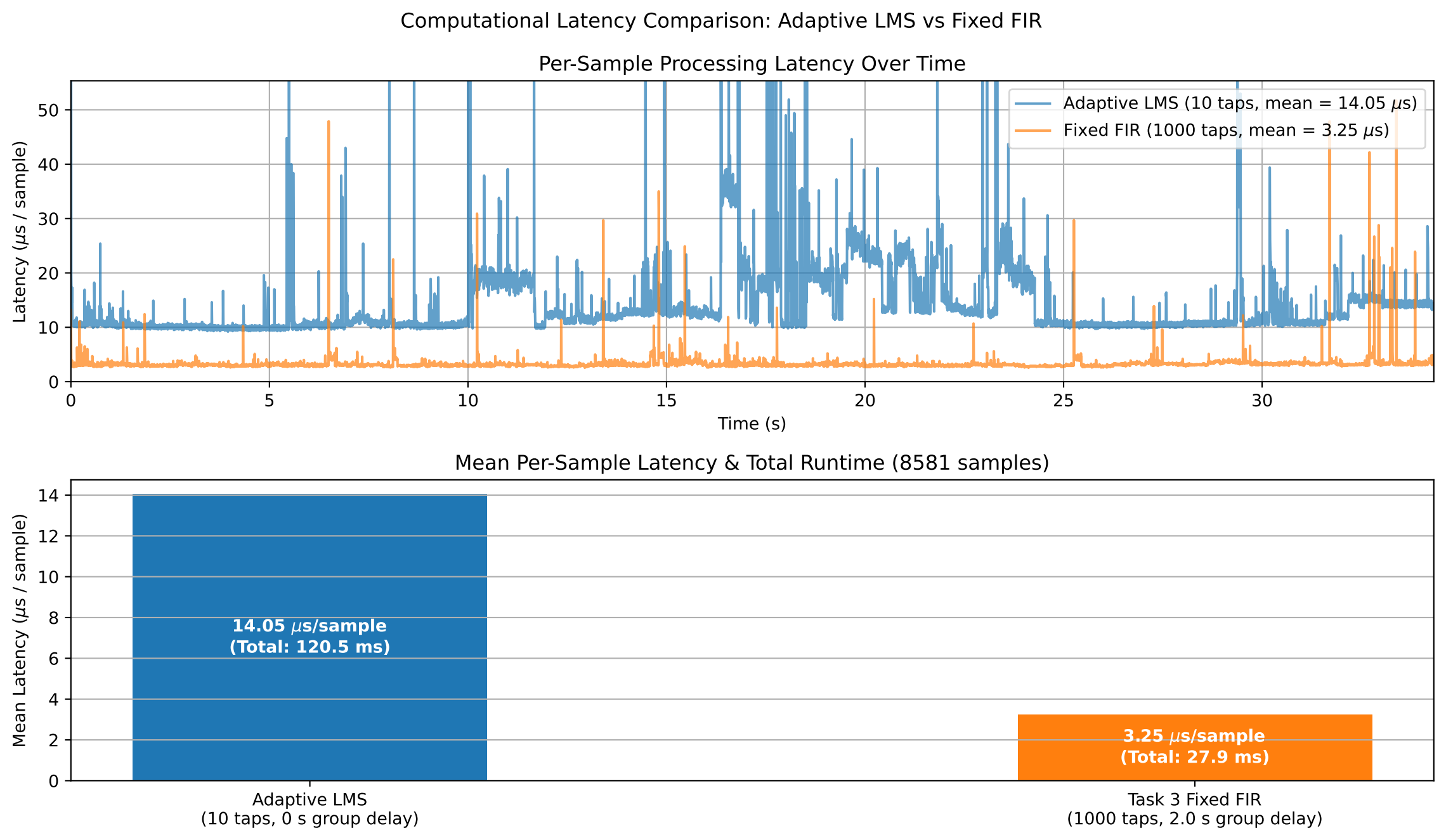

Computational Latency Comparison: Adaptive LMS vs Fixed FIR
Per-Sample Processing Latency Over Time
Adaptive LMS (10 taps, mean = 14.05 μs)
Fixed FIR (1000 taps, mean = 3.25 μs)
Latency (μs / sample)
Time (s)
Mean Per-Sample Latency & Total Runtime (8581 samples)
Mean Latency (μs / sample)
14.05 μs/sample
(Total: 120.5 ms)
3.25 μs/sample
(Total: 27.9 ms)
Adaptive LMS
(10 taps, 0 s group delay)
Task 3 Fixed FIR
(1000 taps, 2.0 s group delay)